电场 $E(t,r) = [\,G(t, fwhm \to \sigma)\; G(r, \frac{w_0}{2})\,]^{1/2}$.

光强 $I(t,r) = \underbrace{G(t, fwhm \to \sigma)}_{\text{时域}}\; \underbrace{G(r, \frac{w_0}{2})}_{\text{空域}}$.

其中, $G(x,\sigma) = \exp[-\frac{1}{2}(\frac{x-x_0}{\sigma})^2]$, $x_0 = 0$.

$fwhm \to \sigma$: $\sigma = \dfrac{fwhm}{2\sqrt{\log 2}}$ (power = 1 case)

$fwhm = \tau$

时域: $I(t) = G(t, fwhm \to \sigma)$

$= \exp[-\frac{1}{2}(t / \frac{\tau}{2\sqrt{2\log 2}})^2]$

$= \exp[-\frac{1}{2}\frac{t^2}{\tau^2} \cdot 8\log 2]$

$= \exp(-4\log 2 \frac{t^2}{\tau^2}) = 2^{-4\frac{t^2}{\tau^2}}$

when $t = \frac{\tau}{2}$, $I(t) = \frac{1}{2} I_0$ ✓

空域: $I(r) = \exp[-\frac{1}{2}(r/\frac{w_0}{2})^2] = \exp(-2\frac{r^2}{w_0^2})$

$E(r) = \exp -\frac{r^2}{w_0^2}$ ✓

当 $r = w_0$ 时, $E(r) = \frac{1}{e} E_0$, w_0 为束腰半径.